

COVENANT

The proof-carrying options agent that tries to break your mandate before it trades.

Declare a mandate. Covenant compiles it into executable policy, fuzz-tests it against thousands of hostile order and portfolio states, then issues a short-lived TradePermit for every order that survives. Alpaca execution is impossible without that permit. Every decision is replayable, and a synchronized shadow book measures what each safety rule saved or cost.

The winning reframe

V1 was a risk-controlled options bot with formal language around it. V2 is a **mandate operating system** with a credible options engine inside it. The distinction matters: deterministic gates are now common in this field. Covenant wins only if it demonstrates a capability competitors do not make tangible - policy compilation, automatic adversarial testing, capability-separated execution, and measured counterfactual governance.

The one-line demo

A judge writes a difficult mandate, Covenant finds a contradiction, repairs it with the judge, attacks the resulting policy, blocks a violating order, issues a valid permit for a legal order, executes it on Alpaca paper, and verifies the complete trace from one command.

What the product promises

Promise	Meaning	Honest boundary
Powerless by construction	The strategy and model have no broker credential or order tool. Only the Permit Executor can call Alpaca.	A broker outage or fill outcome cannot be guaranteed.
Machine-checkable adherence	Every controllable rule is evaluated on a versioned snapshot and recorded in a signed permit and hash-chained event log.	This is replayable evidence, not a universal mathematical proof.
Measured safety	Rejected and downsized candidates enter a no-hindsight shadow ledger using the same timestamps and marks.	The shadow result is a counterfactual estimate, not causal certainty.

Team constraint: three builders, seven days, one alpha engine, one safety kernel, one proof surface. No fourth agent and no general-purpose theorem prover.

What is strong, what is weak, what must be cut

V1 element	Verdict	Why
Plain-English mandate	KEEP	Immediate product value and a strong demo front door.
Deterministic veto/shrink core	KEEP, HARDEN	Necessary, but no longer original. Current teams already advertise hard gates, refusal logs, and powerless-by-construction agents.
LTL formulas	CUT AS HEADLINE	Several formulas are non-standard or too weak. A runtime predicate is not automatically a temporal-logic proof.
Provably never broke the mandate	NARROW	Only controllable actions can be guaranteed. Fills, closures, stale data, and broker availability are external.
60-90 day backtest	REPLACE	Too short for regime claims, IV rank, or option expectancy. Synthetic Black-Scholes chains can create false confidence.
Credit spreads have positive expectancy	DELETE	Unsupported. Short volatility earns premium but can still have negative expectancy after selection, spreads, and jumps.
Flat P&L; reads as disciplined	DELETE	The rubric explicitly includes P&L.; Safety does not substitute for alpha.
Two playbooks across five names	TIGHTEN	Too many degrees of freedom for a seven-day build. Start with SPY/QQQ and one candidate factory.
Three-agent architecture	SIMPLIFY	Agent count is not a judging criterion. Clear authority boundaries matter more.
Both MCP and CLI are required	CORRECT	Current summaries describe Trading API plus either MCP or CLI. Using both can help, but should not be treated as a disqualification rule without organizer confirmation.

The biggest strategic error in V1: it treats the safety gate as the moat. The field already contains Pin Desk with 12 gates and 180+ tests, Vega with 48/48 reproducible claims, Guarded Alpha with deterministic evidence gates, and GammaForge with trade certificates and counterfactuals. Safety alone is table stakes.

Formal claim corrections

- INV-4 uses a non-standard `G_sess` operator. Implement the daily halt as a tested state-machine transition: `ACTIVE -> HALTED`, with reset only at a new market session.
- INV-6 says 'never hold past max DTE' but uses an eventual close. Eventuality has no deadline, and a close fill is not under the agent's control. Guarantee a close attempt and escalation policy, not a fill.
- Portfolio max loss for overlapping option positions is difficult. V2 deliberately sums standalone defined-risk envelopes, a conservative upper bound that is easy to verify.
- A model must never be the source of executable rules. It drafts a typed policy; schema validation, policy templates, contradiction checks, and explicit activation make it live.

Covenant is a self-fuzzing mandate compiler

The product has four components. Only one of them is allowed to reach the broker.

Component	Authority	Output
Mandate Studio	Featherless drafts a typed policy from plain English. It may ask for clarification; it cannot activate policy.	Versioned Policy JSON plus a plain-English echo.
Break Me Engine	Generates hostile candidates and portfolio states, then property-tests every invariant before activation.	Counterexample or passing coverage report.
Alpha Engine	Builds and ranks only pre-approved, defined-risk option structures. No Alpaca order credentials.	TradeIntent containing contracts, limit, quantity, thesis, and expiry.
Safety Kernel + Permit Executor	Re-fetches account and quotes, evaluates policy, signs a narrow TradePermit, and submits only an exact matching order.	Alpaca order ID plus Decision Certificate.

The TradePermit

A permit is a short-lived capability, not an explanation. It binds **mandate_hash**, **account_snapshot_hash**, **market_snapshot_hash**, **exact legs**, **max quantity**, **limit-price band**, **expiration timestamp**, **nonce**, and **intent hash**. The executor rejects missing, expired, replayed, mutated, or over-sized permits. This makes the AI unable to widen a trade after approval.

Judge-friendly proof: change one contract, increase quantity, replay an old permit, or wait past its TTL. All four attacks fail before Alpaca receives an order.

The Break Me Engine

- Converts each policy invariant into property tests over generated orders, quotes, positions, P&L; states, session transitions, and stale-data states.
- Runs a fixed reproducible seed plus randomized seeds. A failing example is minimized and shown as a human-readable counterexample.
- Activation is fail-closed: a policy with an unresolved contradiction, untested rule, or failing property stays in DRAFT.
- Ships with a judge mode: type a mandate, click 'Try to break it,' and watch coverage and counterexamples appear live.

The Shadow Ledger

Every approved, shrunk, vetoed, and abstained candidate receives synchronized marks on the same schedule. The dashboard reports **governed P&L**;; **shadow P&L**;; **safety delta**, **missed-upside cost**, **loss avoided**, and **rule-level attribution**. No outcome is retroactively selected or removed.

Separate intelligence from authority

Stage	Runs	Hard boundary
1. Declare	Plain English -> Featherless typed draft -> schema and contradiction checks -> user activation.	Model has no direct policy write and no broker access.
2. Prove before run	Property-based generator attacks the policy; results are pinned to policy hash and code revision.	A failing or untested policy cannot become ACTIVE.
3. Observe	Alpaca MCP retrieves option chain, quotes, Greeks, account, positions, and clock.	Stale or missing evidence produces ABSTAIN.
4. Build	Deterministic candidate factory generates legal vertical spreads for SPY/QQQ.	No free-form contract symbols from the model.
5. Rank	Scenario engine estimates after-cost payoff distribution and confidence. Featherless may veto or shrink confidence only.	Model multiplier is constrained to [0,1].
6. Permit	Kernel re-fetches state, checks all invariants, signs exact short-TTL permit.	Only kernel owns signing key.
7. Execute	Executor validates permit and submits a multi-leg limit order through Alpaca Trading API.	Only executor owns paper API credentials.
8. Prove after	Hash-chained event log, signed run manifest, replay verifier, and shadow marks.	Reasoning text is evidence context, not execution authority.

Executable invariant set

ID	Invariant	Enforcement
COV-01	Every order exactly matches one unexpired, unused permit.	Executor
COV-02	Every structure has finite standalone max loss within per-trade cap.	Candidate factory + kernel
COV-03	Conservative sum of standalone max losses after trade stays under portfolio heat cap.	Kernel
COV-04	No entry after daily halt until a verified new session.	State machine + executor
COV-05	Quote age, bid/ask width, DTE, delta, and liquidity stay inside mandate bands.	Kernel
COV-06	No duplicate or economically equivalent exposure within the cooldown window.	Kernel + nonce store
COV-07	Exit workflow starts before configured deadline; failed closes escalate and disable new entries.	Position monitor
COV-08	Any missing, stale, inconsistent, or unverified state returns ABSTAIN.	All stages

One command: ``python -m covenant.verify demo/run_manifest.json`` replays every policy decision, validates the hash chain and permit signatures, and reports exact mismatches. Live market facts are clearly labeled RECORDED; deterministic checks are REPRODUCED.

A credible edge, separated from the safety story

The hackathon scores P&L.; Covenant must earn attention as a trader before governance can win the room. The V2 alpha engine is deliberately small: SPY and QQQ, 7-21 DTE, defined-risk verticals, one candidate factory, and one regime selector.

Signal	Implementation	Purpose
Volatility price	Compare chain IV and expected move with an EWMA/range-based realized-volatility forecast.	Distinguish rich from cheap optionality.
Direction	5/20-day trend, intraday momentum, and breadth confirmation with fixed weights.	Choose bullish, bearish, or neutral candidate family.
Scenario distribution	Bootstrap calibrated returns and reprice each candidate payoff after conservative spread/slippage assumptions.	Estimate net expected P&L; and downside.
Confidence gate	Trade only when the lower confidence bound of after-cost expected edge is positive and data quality passes.	Prevent weak point estimates from forcing a trade.

Allowed structures

Regime	Candidate	Why
Directional, IV not cheap	Bull call or bear put debit vertical	Capped premium, directional convexity, lower cost than naked long option.
Range-bound, IV rich	Single-side credit vertical aligned away from trend risk	Defined max loss; simpler and more robust than an iron condor during a short build.
No robust edge	ABSTAIN	No mandate can force the system to invent alpha.

Evidence ladder

- Level 1 - Unit and property tests: payoff, max loss, sizing, quote filters, state transitions, permit tampering, idempotency.
- Level 2 - Two or more years of walk-forward signal validation on underlying data across volatility regimes; no in-sample headline metrics.
- Level 3 - Conservative synthetic option payoff simulation, explicitly labeled SYNTHETIC and never presented as fill-accurate chain history.
- Level 4 - Replay of committed live chain snapshots with bid/ask, Greeks, and exact candidate rankings.
- Level 5 - Fresh \$100,000 Alpaca paper account with real multi-leg order IDs, fills, and competition-window P&L;

Default demo risk

Profile	Per trade	Portfolio heat	Daily halt	Model authority
Conservative	0.35%	1.50%	-0.75%	Veto or shrink
Balanced	0.60%	2.50%	-1.25%	Veto or shrink
Aggressive	1.00%	4.00%	-2.00%	Veto or shrink

Risk limits are examples for paper trading, not investment recommendations.

Judge the product in 90 seconds

Criterion	Covenant V2 evidence	Judge takeaway
P&L; performance	Live competition account, after-cost candidate ranking, walk-forward validation, and a compact two-underlying universe.	The team has an actual alpha process, not only safety theater.
Technology implementation	Trading API multi-leg orders; Alpaca MCP for chains, Greeks, account, and clock; official CLI/replay commands; Featherless for typed mandate drafting and adversarial critique.	Every sponsor surface has a clear job.
Creativity and originality	Self-fuzzing mandates, signed short-lived TradePermits, judge-authored attack mode, and rule-attributed shadow P&L;	The agent proves how it is constrained and what those constraints changed.
Presentation and execution	One live causal sequence from mandate to attack to veto to permit to Alpaca order to replay verification.	No architecture-tour demo and no hidden manual step.
Social engagement	Public 'Can you break our trader?' challenge with daily counterexample, fix, and verified paper result.	The audience participates instead of reading build logs.

90-second demo script

Time	Screen	Action
0-12s	Mandate Studio	Judge types: 'Grow the account, max 0.5% per trade, no naked options, stop at -1%, but trade at least once daily.' Compiler flags the contradiction between forced trading and abstention safety.
12-25s	Break Me	Activate repaired policy and run hostile tests. Show minimized attempts for over-size, stale quote, replayed permit, and duplicate exposure.
25-42s	Live candidate	Agent ranks legal SPY/QQQ spreads; model may veto or reduce confidence, never widen.
42-56s	Attack console	Mutate quantity after approval. Executor rejects the altered order because intent hash and permit no longer match.
56-70s	Alpaca execution	Issue a fresh permit, submit exact multi-leg paper order, display Alpaca order ID and status.
70-82s	Proof Explorer	Run verifier: signature valid, chain valid, policy checks reproduced, recorded facts labeled.
82-90s	Shadow Ledger	Show governed vs shadow result and close on live competition P&L; plus safety delta.

Closing line: 'Most agents explain why they traded. Covenant proves what they were allowed to do, tries to break that authority, and measures what the boundary changed.'

Build the irreversible demo path first

Day	Must ship	Owner split	Exit test
1	Fresh \$100k paper account; Level 3 options; one manual multi-leg order; frozen Policy JSON and TradeIntent schemas.	All	Real order ID and account ID captured.
2	SPY/QQQ chain ingestion; candidate payoff/max-loss engine; quote/liquidity filters; deterministic sizing.	Alpha lead	Recorded chain -> ranked legal candidates.
3	Safety kernel, session state machine, permit schema/signature/TTL/nonce, executor credential isolation.	Safety + backend	Four permit attacks fail before API call.
4	Autonomous loop on paper; order monitoring; close workflow; hash-chained event journal.	Alpha + safety	Scheduled run places or abstains without manual code changes.
5	Mandate Studio; Featherless typed draft; contradiction checks; property-based Break Me suite.	Product + safety	New mandate generates passing tests or a minimized counterexample.
6	Proof Explorer, one-command verifier, Shadow Ledger, deployed dashboard.	Frontend + all	Fresh machine reproduces deterministic checks.
7	Walk-forward report, README, 90-second video, one-page write-up, submission, social challenge.	All	Cold judge run succeeds twice.

Scope fence

Build	Do not build
One constrained policy schema with 8 executable invariants	General natural-language policy language or theorem prover
SPY and QQQ vertical spreads	Five-symbol universe, naked options, calendars, condors, or assignment-heavy structures
Property tests plus deterministic replay	Blockchain, zero-knowledge proofs, or unverifiable 'AI proof' language
One model with three narrow roles: draft, veto, explain	Agent council, debate swarm, self-modifying strategy, or fourth agent
One deployed proof surface	Separate dashboards for backtest, trading, and risk

Kill criteria

- If Level 3 multi-leg paper execution is not confirmed by noon Day 1, use one long debit vertical only and remove credit strategy claims.
- If permit isolation is not demonstrable by Day 3, cut 'powerless by construction' and call it a deterministic pre-trade gate.
- If Break Me cannot generate a real counterexample by Day 5, make it a fixed adversarial conformance suite and stop calling it self-fuzzing.
- If shadow marks are incomplete, show candidate history without P&L; attribution. Never backfill favorable outcomes.
- If no live paper order exists by Day 4, stop frontend work until the execution path is real.

What must be true before submit

Area	Non-negotiable evidence
Alpaca	Dedicated competition paper account, \$100,000 start, Level 3 confirmed, account ID included, real multi-leg order IDs, paper-only endpoint allowlist.
Agent	Autonomous scheduled loop; clear AI role; strategy cannot access broker credentials; failure returns ABSTAIN.
Options	Defined-risk verticals, exact max-loss calculation, fresh quotes, spread and liquidity checks, position and exit monitoring.
Mandate	Versioned policy, plain-English echo, explicit activation, contradiction check, 8 executable invariants, property coverage report.
Permit	Signed, short TTL, exact intent binding, nonce and replay prevention, quantity and price limits, tamper demo.
Proof	Hash-chained events, signed run manifest, one-command replay, recorded vs reproduced labels, at least one public self-correction.
Performance	Competition P&L;, walk-forward signal evidence, conservative costs, no unlabelled synthetic option history.
Presentation	Live app, public repository, setup in under 10 minutes, 90-second video, one-page write-up, deck, cold demo twice.

Submission copy

Covenant is a proof-carrying options agent built on Alpaca. A trader declares a mandate in plain English; Covenant compiles it into executable policy, attacks that policy with generated hostile states, and refuses activation until every safety property passes. Its alpha engine ranks defined-risk SPY and QQQ spreads, but cannot submit orders. Only a separate executor can reach Alpaca, and only with a signed, short-lived TradePermit bound to the exact contracts, quantity, price band, account state, and mandate version. Every approved, shrunk, vetoed, and abstained decision is replayable, while a synchronized shadow ledger measures the P&L; saved or sacrificed by each rule. The result is not an agent that asks to be trusted. It shows its authority, attempts to break it, and proves every broker action matched it.

Sources checked for this re-lock

- lablab.ai Alpaca AI Trading Agents Hackathon event and live dashboard, accessed 29 Aug 2026.
- Alpaca Trading MCP Server documentation, including option chains, Greeks, multi-leg spreads, toolset filtering, and paper defaults.
- Alpaca multi-leg options paper-trading documentation and July 2026 options API tutorial.
- Current public submissions and team pages, including Pin Desk, Vega, AlphaPilot, VibeHedge, Guarded Alpha Lab, GammaForge AI, tradesense, and others, accessed 29 Aug 2026.

Decision: keep the name Covenant. Retire Bellatrix as a public-facing brand. Call the enforcement layer the Safety Kernel, the signed artifact a TradePermit, the pre-deployment attack surface Break Me, and the audit view Proof Explorer.

Paper trading only. Not investment advice. Options involve significant risk. Paper results do not represent live execution and may omit slippage, liquidity constraints, or delays.